Methods of Theoretical Physics

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Abstract

This course was taught by Dr. Claudio Chamon during the Spring 2025 semester at Boston University. We used the text Basic Training in Mathematics: A Fitness Program for Science Students by R. Shankar.

Contents

Chapter 1

Sets

Lecture 1: Syllabus Day

Tue 21 Jan 2025 12:30

The professor is recommending ChatGPT as a tutor. This is awesome. Problem sets due Thursdays at 11:59pm.

Prof asked "What are numbers?" and this chick answered "erm uh numebrs are uh do you want me to give the set theory definition? Erm its difficult to define numbers it requires complex set theory" lmfao. We will define numbers as an element of an algebraic structure, like a field or group. We take $0 \in \mathbb{N}$. We use the convention of writing $\mathbb{Z}_2$ instead of $\mathbb{Z}/2\mathbb{Z}$.

A cartesian product is a product in Set; an object $A \times B$ together with canonical projections that are final in the category $\mathsf{Set}^{A,B}$.

Lecture 2: lol i should drop this class

Thu 23 Jan 2025 12:31

we learned python.

Lecture 3: Cauchy-Riemann

Tue 11 Feb 2025 12:30

There are a few Cauchy-Riemann conditions. I think we are viewing a complex function $f(x+iy)$ as a real function $f(x,y)$.

- $$i\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}.$$
- $f(x,y) = u(x,y) + iv(x,y)$.
- $$-\frac{\partial v}{\partial x} = \frac{\partial u}{\partial x}.$$
- $$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}.$$

If this is true, this means f is a function of z only and not z^*. If these conditions are satisfied, $f'(z)$ exists.

Definition 1.1: Analytic

A function $f(z)$ is analytic in an open region $R \subseteq \mathbb{C}$ if and only if $f'(z)$ exists in R. f is entire if $R = \mathbb{C}$.

For example, $f(z) = z^2$ is entire because $f'(z) = 2z$. Similarly, $f(z) = e^z$ is entire because $f'(z) = f(z)$. Given an analytic function, we can talk about singularities.

Definition 1.2: Singularity

A singularity of an analytic function is an isolated point z such that f is not analytic at z, but is analytic in an open disc surrounding z.

For example, $f(z) = 1/z$ has a singularity at $z = 0$. We often call this a *pole*. Similarly, $f(z) = 1/z^2$ has a singularity at $z = 0$ as well. More generally,

$$f(z) = \frac{R(z_0)}{z - z_0}$$

has a singularity at z_0. We call $R(z_0)$ the *residue* of the pole, and z_0 is the location of the pole.

For example, the residue of e^z/z is

$$\lim_{z \to 0} z f(z) = 1.$$

For example,

$$f(z) = \frac{1}{z+i} = \frac{1}{z-i} = \frac{2z}{z^2+1}$$

has poles at $\pm i$.

Poles generalize to higher order poles. Let

$$f(z) = \frac{C}{(z - z_0)^n}.$$

This function has a pole at z_0 of order n, with residue C.

Another type of singularity is an essential singularity. Consider $f(z) = e^{1/z}$. The taylor series of this is

$$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{1}{z^n}.$$

Note that

$$\lim_{z \to 0} z^p f(z)$$

is not finite for all $p \in \mathbb{N}$.

The final type of singularity we will discuss is the branch cut. For example, consider $f(z) = \sqrt{z}$. There is some ambiguity in the square root, so branch cuts will sort this out. If we write

$$z = \exp i\theta,$$

then

$$z^{1/2} = \exp(i\theta/2).$$

However, we could have also written

$$z = \exp(i\,(\theta + 2\pi)).$$

We require this phase angle to be the same, otherwise we have a branch cut.

A contour integral is an integral from z_1 to z_2, where we integrate along a path P from z_1 to z_2. That is, we take the integral

$$\int_P f(z)\, \mathrm{d}z.$$

Computing this is the same as a multivariate limit; just parameterize P with a function $z(t) = x(t) + iy(t)$ for $t \in [0, 1]$, and then take the integral

$$\int_0^1 f(z(t)) \frac{\mathrm{d}z}{\mathrm{d}t} \, \mathrm{d}t.$$

Path independence is when two distinct paths P_1, P_2 have the same integral

$$\int_{P_1} f(z) \, \mathrm{d}z = \int_{P_2} f(z) \, \mathrm{d}z.$$